

Shivam Rathod

AI Engineer - Generative AI & Intelligent Systems

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PROFESSIONAL SUMMARY

AI Engineer building production-ready inference pipelines, RAG systems, and semantic search architectures. Focused on reducing inference latency, optimizing retrieval workflows, and deploying robust NLP/Computer Vision systems using FastAPI, TensorFlow, and vector databases.

EXPERIENCE

AI Developer (Freelance)

Remote

Resume Forge - AI-Powered Optimization Platform

Oct 2025 - Nov 2025

- Engineered a semantic ATS matching pipeline using **KeyBERT** and **SentenceTransformers (SBERT)**, extracting up to **15 key skills** per job description to calculate highly accurate similarity scores.
- Architected a **hybrid AI evaluation pipeline** and automated PDF generator, processing embeddings locally before invoking Gemini to reduce latency to **1-4 seconds** and minimize LLM payload dependency.

Backend & Automation Developer (Freelance)

Remote

FieldLens - Field Data Verification Platform

Sept 2025 - Oct 2025

- Developed an automation-first backend platform for telecom field verification, orchestrating real-time image validation, blur detection, and automated MongoDB synchronization to reduce manual validation delays by **80%**.
- Built a bilingual **Twilio WhatsApp automation system** sequentially guiding technicians through a strict **14-step verification workflow**, accelerating standardized field data collection across operations.

PROJECTS

- Codebase Intelligence - AI-Powered Repository Analysis Platform** [GitHub](#) | [Live](#)
Built a repository intelligence platform supporting Python, JavaScript, and TypeScript repositories, integrating interactive architecture visualization to transform GitHub repositories into searchable knowledge bases.
Orchestrated **RAG pipelines, vector embeddings, and Qdrant retrieval** using FastAPI and React to parse AST-based dependency graphs, enabling natural-language querying of raw source code.
- KrishiMitra AI - Multimodal Agricultural Intelligence System** [Live](#) | [GitHub](#) | [Demo](#)
Engineered a hybrid crop disease diagnosis pipeline covering 27 disease categories using EfficientNetB0 and Gemini Vision, evaluating PlantVillage (96.3% validation accuracy) and PlantDoc datasets to prioritize real-world deployment robustness.
Built a multimodal agricultural assistant using FastAPI and Twilio WhatsApp Webhooks, enabling image-based disease diagnosis through confidence-aware computer vision inference and automated response generation.

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI & GenAI: TensorFlow, NLP, Computer Vision, Semantic Search, RAG, AI Agents, LangChain, Sentence Transformers, Qdrant, LLM APIs

Backend: FastAPI, Django, REST APIs, Supabase, Cloudinary

Tools: Git, GitHub, Docker, Hugging Face Spaces, Streamlit

EDUCATION

International Institute of Professional Studies, DAVV

Indore, India

Integrated M.Tech (Information Technology) - **8.80 CGPA**

Sept 2023 - Jun 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Architecture.

ACHIEVEMENTS

- Top 8 - IBM SkillsBuild PBL (AI Agent Architect, Sept 2025)** for *KrishiMitra-AI*.
- Published open-source AI agent architectures utilizing LangChain and Ollama, earning **14+ GitHub stars**. [🔗](#)
- Achieved a peak **LeetCode Contest Rating of 1565**, placing in the **Top 30%** globally among active competitors.
- Solved **650+** DSA and competitive programming challenges across multiple platforms. [🔗](#)
- Honorable Mention - DC Hackathon 2025** for engineering the *Perceptron Visualizer App*.
- Competed in the **NASA Space Apps Challenge, OpenAI x NxtWave Hackathon, and SIH**, collaborating on rapid technical problem-solving.

CERTIFICATIONS

- Generative AI Mastery Workshop** - OpenAI Academy x NxtWave
- AI Agent Architect PBL Program** - IBM SkillsBuild
- Cloud Computing & GenAI Foundations** - Google Cloud